

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Aashika S P	Reg. No.:	192521068
Date:	1.10.2026	Duration:	60 minutes

Code of Conduct

I Aashika S P (192521068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
- 6. Discuss Expected Outcomes and Limitations**
- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing;

	technical justification.			report lacks organization and clarity.
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Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Honey Chain: A Blockchain-Based System for Honey Traceability and Smart Beekeeping Management

1. Understand the Problem Statement

Explain the problem in your own words.

The honey industry involves several stages, including beekeeping, honey collection, processing, quality testing, packaging, transportation, and selling. At present, information about these stages may be recorded manually or stored in separate systems. This makes it difficult to verify the origin, quality, processing history, and transportation details of honey.

*The proposed **Honey Chain** system uses blockchain technology to create a transparent and tamper-resistant record of honey throughout its lifecycle. Beekeepers can record information about bee colonies and honey harvesting, while processors, quality inspectors, distributors, retailers, and consumers can access appropriate information about the product.*

Identify the objectives of the proposed system.

The main objectives of the Honey Chain system are:

- *To provide end-to-end traceability of honey from the apiary to the consumer.*
- *To securely store honey production and supply-chain information using blockchain.*
- *To improve transparency and trust among stakeholders.*
- *To maintain records of honey harvesting, processing, quality testing, packaging, and transportation.*
- *To support smart beekeeping through monitoring of hive-related information.*
- *To allow consumers to verify the origin and history of honey using a QR code.*
- *To reduce the possibility of false or modified product information.*
- *To improve the efficiency of honey supply-chain management.*

State the scope of the problem.

The scope of Honey Chain covers the complete honey supply chain, beginning with beekeepers and bee colonies and continuing through harvesting, processing, quality verification, packaging, distribution, retail, and final consumer verification.

The system can also be extended with IoT sensors to collect information such as hive temperature, humidity, weight, and environmental conditions. Blockchain can then be used to maintain trustworthy records of important events. The system is mainly focused on traceability, transparency, data integrity, and smart management rather than replacing physical laboratory testing or regulatory certification.

2. Identify Key Stakeholders and Challenges

Identify all the stakeholders involved in the system.

The major stakeholders of the Honey Chain system are:

1. **Beekeepers** – Manage bee colonies, maintain hives, and collect honey.
2. **Honey Processors** – Process, filter, test, and package collected honey.
3. **Quality Inspectors/Laboratories** – Verify the quality and safety of honey.
4. **Distributors** – Transport honey products between processors, warehouses, retailers, and other locations.
5. **Retailers** – Sell verified honey products to consumers.
6. **Consumers** – Check the origin, quality information, and journey of honey.
7. **System Administrator** – Manages users, permissions, system configuration, and security.

8. **Regulatory Authorities** – May verify compliance, certifications, and required standards.
9. **IoT/Technology Providers** – Provide sensors and supporting technologies for smart hive monitoring.

Explain the roles and responsibilities of each stakeholder.

Beekeepers are responsible for registering their apiaries, maintaining hive information, recording harvesting details, and updating relevant production information. If IoT devices are used, they can also monitor environmental and hive conditions.

Processors receive harvested honey and record processing activities such as filtering, batch creation, packaging, and storage. They are responsible for maintaining accurate batch information.

Quality inspectors and laboratories perform quality checks and record test results associated with a particular honey batch. These records provide additional confidence about the product.

Distributors and retailers update transportation, storage, and product movement information. Consumers can use the final product's QR code to view selected traceability information.

Consumers are the end users of the traceability system. They can scan a QR code and verify information such as the honey source, harvest date, batch number, processing information, and quality verification.

Discuss the major challenges and issues faced by the stakeholders.

The stakeholders may face several challenges. Beekeepers may have limited access to digital systems and may find manual data entry time-consuming. Processors and distributors may maintain information using separate systems, creating difficulties in tracking a product across the entire supply chain.

Another major challenge is data trust. If records are stored in a centralized or paper-based system, unauthorized modification or loss of information can occur. Consumers also have limited visibility into the actual origin and journey of honey. Therefore, a common system is required to improve data integrity, transparency, and traceability.

3. Analyze the Current Approach

Describe the existing system or current process.

In a traditional honey supply chain, information is commonly recorded using paper documents, spreadsheets, invoices, labels, or independent databases. A beekeeper records harvesting information, after which honey is transferred to a processor. The processor may maintain separate records for processing and packaging. Distributors and retailers then maintain their own transportation and sales records.

Because these records are often maintained separately, connecting the complete history of a particular honey batch can be difficult. Consumers generally receive only limited information from the product label.

Identify its strengths and limitations.

The existing approach is simple and familiar to stakeholders. Small-scale beekeepers can maintain records without requiring advanced technology, and paper or spreadsheet-based systems can be inexpensive to operate.

However, the approach has significant limitations. Records may be lost, duplicated, incorrectly entered, or modified without an easily verifiable history. Information is distributed among different stakeholders, making end-to-end traceability difficult. Consumers may also be unable to independently verify whether the information provided about the honey is accurate. Analyze the problems or gaps that need to be addressed.

The major gap is the absence of a unified and trustworthy traceability mechanism. There is no easy way to connect information about the beekeeper, hive, harvesting, processing, quality testing, packaging, and distribution into one verifiable history.

Honey Chain addresses this gap by assigning a unique identity to each honey batch and recording important supply-chain events on a blockchain network. A QR code can connect the physical product with its digital traceability record.

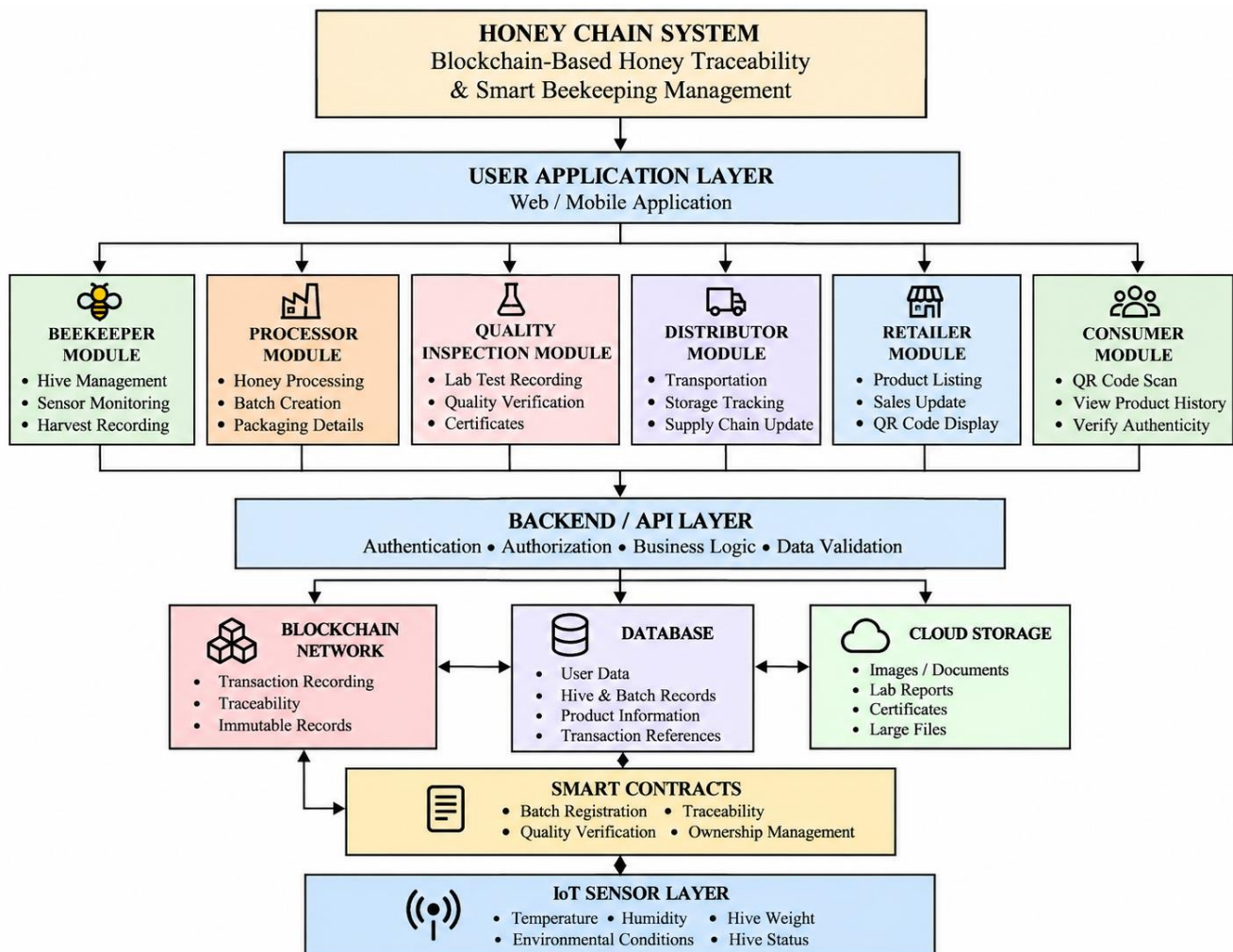
4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

The proposed **Honey Chain** is a blockchain-based software platform that records important events throughout the honey supply chain. Each honey batch receives a unique batch ID, and authorized stakeholders can add relevant information to the system.

The architecture combines a user application, backend services, blockchain, database, IoT devices, and QR-based consumer access. Sensitive or large data can be stored in an appropriate database or cloud storage, while important verification records and transaction references can be recorded on the blockchain.

Architecture Block



Simple data flow

Bee Hive



IoT Sensors



Beekeeper



Honey Harvesting



Unique Honey Batch ID



Processing & Quality Testing



Blockchain Record



Packaging & Distribution



QR Code



Consumer



Verify Honey Origin & History

Describe the major functional modules of the proposed system.

1. User Registration and Authentication Module:

This module allows beekeepers, processors, inspectors, distributors, retailers, and administrators to securely register and log in. Role-based access control ensures that users can access only the functions relevant to their responsibilities.

2. Beekeeper and Hive Management Module:

Beekeepers can register apiaries and hives and maintain information about bee colonies. IoT sensors can optionally provide information such as temperature, humidity, and hive weight.

3. Honey Harvesting Module:

The beekeeper can create a unique honey batch after harvesting. Details such as harvest date, apiary location, hive ID, quantity, and batch number can be recorded.

4. Processing and Packaging Module:

Processors can update information about filtering, processing, quality checks, packaging, and storage. Each activity can be associated with the corresponding batch ID.

5. Quality Verification Module:

Quality inspectors or laboratories can record test information for a honey batch. The system can associate the test result with the unique batch identity.

6. Blockchain Traceability Module:

Important transactions are recorded on the blockchain. Since blockchain records are designed to be tamper-evident, stakeholders can verify the sequence of events associated with a batch.

7. Distribution and Supply Chain Module:

This module tracks the movement of honey from processing facilities to warehouses, distributors, and retailers.

8. QR Code Verification Module:

A unique QR code can be generated for each final product or batch. Consumers can scan the QR code to view authorized traceability information.

9. Admin and Monitoring Module:

The administrator can manage users, permissions, system activity, and abnormal transactions while maintaining appropriate security controls.

Identify the functional and non-functional requirements.

Functional Requirements:

- *User registration and login.*
- *Role-based access control.*
- *Apiary and hive registration.*
- *Honey batch creation.*
- *Harvest data recording.*
- *Processing and packaging record management.*
- *Quality test record management.*
- *Blockchain transaction recording.*
- *Supply-chain tracking.*
- *QR code generation and scanning.*
- *Consumer traceability verification.*
- *IoT sensor data integration.*
- *Report generation.*
- *Administrator management.*

Non-Functional Requirements:

- **Security:** User authentication, authorization, encryption, and secure communication should be implemented.
- **Reliability:** The system should maintain accurate records and minimize service failures.
- **Scalability:** The architecture should support increasing numbers of beekeepers, batches, transactions, and consumers.
- **Usability:** Interfaces should be simple enough for stakeholders with different levels of technical knowledge.
- **Maintainability:** The system should use modular components so individual modules can be updated without affecting the complete application.
- **Performance:** Product information and QR verification should be retrieved within an acceptable response time.
- **Availability:** The system should remain accessible to authorized users when required.
- **Data Integrity:** Important records should be protected from unauthorized modification.

Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

Modularity is achieved by dividing the system into independent modules such as user management, hive management, harvesting, processing, blockchain, quality verification, and QR verification.

Scalability is supported through a layered architecture and cloud-based deployment. Additional users, apiaries, products, and transactions can be accommodated as the system grows.

Maintainability is achieved through well-defined APIs, modular source code, documentation, testing, and separation of frontend, backend, blockchain, and database components.

Reliability is supported through validation, transaction verification, backup mechanisms, error handling, and blockchain-based record integrity.

Usability is incorporated through simple dashboards, clear forms, mobile-friendly interfaces, and QR-code-based consumer access.

Security is implemented through authentication, role-based authorization, encrypted communication, secure API design, input validation, and controlled access to blockchain and database records.

5. Justify the Chosen Methodology/Framework

Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

The **Agile methodology with Scrum** is suitable for developing the Honey Chain system. Scrum divides development into short iterations called sprints. Each sprint can focus on a particular group of features, such as user authentication, hive management, blockchain integration, QR verification, or IoT integration.

Justify why the selected methodology/framework is suitable for the proposed system.

Honey Chain contains multiple technologies and stakeholders, including blockchain, web/mobile applications, databases, QR codes, and potentially IoT devices. Requirements may change during development as stakeholders provide feedback.

Agile Scrum allows the development team to create a basic working version first and gradually add advanced functionality. For example, the first sprint can implement registration and authentication, the next sprint can implement honey batch tracking, and later sprints can integrate blockchain and QR-based verification.

Explain how it supports successful project development and maintenance.

Scrum provides regular planning, development, testing, review, and feedback. At the end of each sprint, stakeholders can review the developed functionality and suggest changes.

Continuous testing helps identify defects early. After deployment, the same iterative approach can be used to introduce new features such as advanced IoT monitoring, analytics, mobile applications, and additional supply-chain integrations. Therefore, Agile Scrum supports both initial development and long-term maintenance of Honey Chain.

6. Discuss Expected Outcomes and Limitations

Describe the expected benefits of the proposed solution.

The Honey Chain system is expected to improve transparency and traceability throughout the honey supply chain. Stakeholders can obtain a connected digital history of a honey batch instead of depending entirely on separate manual records.

Beekeepers can maintain organized production records, processors can track batch information, quality inspectors can associate test results with specific batches, and consumers can verify product information using QR codes.

Blockchain can also provide a tamper-evident record of important supply-chain events, increasing confidence in the integrity of recorded information.

Explain how the solution addresses the identified challenges.

The proposed system addresses fragmented record keeping by connecting important supply-chain events through a unique batch ID. It addresses traceability problems by maintaining a chronological digital history of the product.

The QR verification mechanism provides consumers with easier access to product information. Role-based access and authentication help prevent unauthorized users from modifying or accessing restricted information. IoT integration can further reduce manual monitoring by automatically collecting hive-related environmental data.

Discuss potential risks, implementation challenges, and limitations.

The implementation of Honey Chain may face several challenges. Small-scale beekeepers may require training and simple interfaces to use the system effectively. Internet connectivity may also be a limitation in rural apiary locations.

Blockchain implementation can introduce additional technical complexity, transaction costs, and scalability considerations depending on the selected blockchain platform. IoT sensors may require regular maintenance, calibration, battery replacement, and reliable network connectivity.

Another important limitation is that blockchain can protect the integrity of recorded data after it is entered, but it cannot automatically guarantee that the original information is truthful. For example, if incorrect harvesting information is entered into the system, blockchain will not independently determine that the information is false. Therefore, physical inspections, trusted data-entry procedures, laboratory testing, and appropriate verification mechanisms remain important.

Suggest possible future enhancements.

Future versions of Honey Chain can include:

- *AI-based prediction of hive health and honey production.*
- *IoT-based real-time hive monitoring.*
- *Mobile applications for beekeepers.*
- *GPS-based apiary and transportation tracking.*
- *Automated alerts for abnormal hive conditions.*
- *Advanced honey quality analytics.*
- *Digital certification and regulatory integration.*
- *Multi-language support for local beekeepers.*
- *Integration with smart contracts for automated supply-chain transactions.*
- *Consumer analytics and product authenticity verification.*
- *Integration with weather and environmental data.*
- *Predictive maintenance for IoT devices.*

*Overall, **Honey Chain** can provide a software-based framework for connecting beekeeping and honey supply-chain activities through secure digital records, blockchain-based traceability, and smart monitoring technologies. The proposed system aims to improve transparency, data integrity, and accessibility while providing a foundation for future intelligent beekeeping applications.*